

JESS Thermodynamic Database v8.9

Polonium

24-Jan-23

Reaction No. 9383							
$\text{PoO}+2 + 2\text{<H+1_NTA-3>} = \text{Po}+4_ \text{NTA-3(2)} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:8.18 (3SF)	0	MDS	1118
2	25	0.4	NaClO4	lgK:5.18 (3SF)	3	MIX	1118

Reaction No. 42861							
$2\text{<Oxalic-2>} + \text{PoO}+2 = \text{PoO}+2_ \text{Oxalic-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:7.78 (3SF)	4	MDS	906
2	22	1	NaClO4	lgK:7.78 (3SF)	3	MDS	22553
3	22	1	NaClO4	lgK:7.48 (3SF)	3	MIX	22553
4	25	0.4	Unknown	lgK:7.48 (3SF)	4	MIX	906

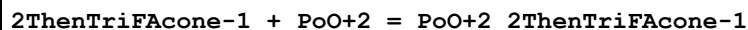
Reaction No. 43263							
$2\text{<Tartaric-2>} + \text{PoO}+2 = \text{PoO}+2_ \text{Tartaric-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:7.30 (3SF)	4	MDS	906
2	22	1	NaClO4	lgK:7.30 (3SF)	3	MDS	22553
3	25	0.4	NaClO4	lgK:7.90 (3SF)	4	MIX	906

Reaction No. 43743							
$\text{PoO}+2 + 2\text{<H+1_Citric-3>} = \text{PoO}+2_ \text{H+1(2)_Citric-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:9.04 (3SF)	3	MDS	906
2	22	1	NaClO4	lgK:9.04 (3SF)	3	MDS	22553
3	25	0.4	Unknown	lgK:6.85 (3SF)	0	MIX	906
Note (3): Low wgt for internal consistency							

Reaction No. 44016							
$\text{PoO}+2 + 2\text{<H+1_NTA-3>} = \text{PoO}+2_ \text{H+1(2)_NTA-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

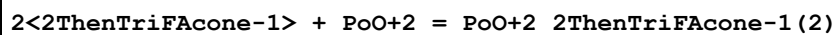
1	22	1	NaClO ₄	lgK:5.78 (3SF)	0	MDS	906
2	22	1	NaClO ₄	lgK:8.18 (3SF)	3	MDS	906
3	22	1	NaClO ₄	lgK:8.18 (3SF)	3	MDS	22553
4	25	0	Inf. Dilution	lgK:8.2 (2SF)	1	ESC	

Reaction No. 44569



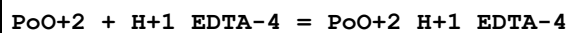
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO ₄	lgK:7.60 (3SF)	1	MDS	906
Note (1): Surely refers to same datum; difficult to know which in error							
2	22	1	NaClO ₄	lgK:6.60 (3SF)	1	MDS	22553

Reaction No. 44570



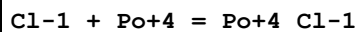
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO ₄	lgK:13.11 (4SF)	3	MDS	906
2	22	1	NaClO ₄	lgK:13.11 (4SF)	3	MDS	22553

Reaction No. 44870



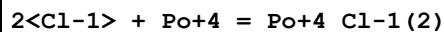
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO ₄	lgK:8.00 (3SF)	3	MDS	906
2	22	1	NaClO ₄	lgK:8.00 (3SF)	3	MDS	22553
3	25	0.4	Unknown	lgK:5.95 (3SF)	0	MIX	906
Note (3): Low wgt for internal consistency							

Reaction No. 54268



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22:25	1	HClO ₄	lgK:2.34 (3SF)	5	MCE	931
2	23	1	H(Cl,ClO ₄)	lgK:2.34 (3SF)	3	MDS	22556
3	23	4:6	HCl	lgK:2.56 (3SF)	4	MDS	931
4	23	4:6	HCl	lgK:2.56 (3SF)	3	MDS	22556

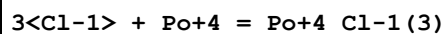
Reaction No. 54269



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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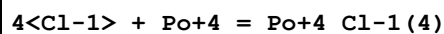
1	22:25	1	HClO4	lgK:4.42 (3SF)	5	MCE	931
2	23	1	H (Cl,ClO4)	lgK:4.41 (3SF)	3	MDS	22556
3	23	4:6	HCl	lgK:4.80 (3SF)	4	MDS	931
4	23	4:6	HCl	lgK:4.80 (3SF)	3	MDS	22556

Reaction No. 54270



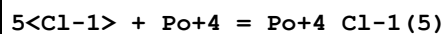
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22:25	1	HClO4	lgK:6.34 (3SF)	5	MCE	931
2	23	1	H (Cl,ClO4)	lgK:6.34 (3SF)	3	MDS	22556
3	23	4:6	HCl	lgK:6.88 (3SF)	4	MDS	931
4	23	4:6	HCl	lgK:6.88 (3SF)	3	MDS	22556

Reaction No. 54271



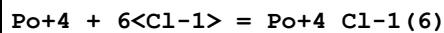
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22:25	1	HClO4	lgK:8.53 (3SF)	5	MCE	931
2	23	1	H (Cl,ClO4)	lgK:8.53 (3SF)	3	MDS	22556
3	23	4:6	HCl	lgK:8.85 (3SF)	4	MDS	931
4	23	4:6	HCl	lgK:8.85 (3SF)	3	MDS	22556

Reaction No. 54272



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22:25	1	HClO4	lgK:10.08 (4SF)	5	MCE	931
2	23	1	H (Cl,ClO4)	lgK:10.08 (4SF)	3	MDS	22556
3	23	4:6	HCl	lgK:10.60 (4SF)	4	MDS	931
4	23	4:6	HCl	lgK:10.60 (4SF)	3	MDS	22556

Reaction No. 54273



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22:25	1	HClO4	lgK:11.57 (4SF)	5	MCE	931
2	23	0	Unknown	lgK:14. (2SF)	0	MEF	22537
3	23	1	H (Cl,ClO4)	lgK:11.57 (4SF)	3	MDS	22556
4	25	0	Inf. Dilution	lgK:7.8 (2SF)	5	CRV	31246

Note (4): Table 2, p. 28

5	22	1	HCl	lgR:14. (2SF)	3	MEF	22542
6	23	4:6	HCl	lgR:11.92 (4SF)	4	MDS	931
7	23	4:6	HCl	lgR:11.92 (4SF)	3	MDS	22556
8	25	0	Inf. Dilution	dG:-44.5 (3SF) kJ	5	CRV	31246

Reaction No. 54274							
Po+4_C1-1(5) + Cl-1 = Po+4_C1-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:2.3 (2SF)	0	MDS	931
2	25	0	Inf. Dilution	lgK:-5.3 (2SF)	1	ELC	

Reaction No. 71038							
Po+4 + 4<e-1> = Po(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.6 (3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:50.2 (3SF)	0	EOD	
Note (2): Personal communication FMEO							
3	25	0	Inf. Dilution	dG:293. (3SF) kJ	0	CRV	3006
4	25	0	Inf. Dilution	dG:-276.5 (4SF) kJ	3	CRV	31246
Note (4): lgK "very poorly defined"; see also Table 2, p. 28							
5	22	1	HNO3	E:0.76 (2SF)	0	MEF	22542
6	25	0	Inf. Dilution	E:0.87 (2SF)	0	CRV	17631
Note (6): See p. 26. Gives +0.6 wrt calomel electrode (+0.27)							
7	25	0	Inf. Dilution	E:0.76 (2SF)	0	CRV	22519
Note (7): States Po+4 but almost certainly refers to PoO+2 = Po(OH)2+2							
8	25	0	Inf. Dilution	E:0.765 (3SF)	0	MEF	22521
9	25	0	Inf. Dilution	E:0.76 (2SF)	0	MEF	22523

Reaction No. 71039							
Po+4 + 2<e-1> = Po+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:27.5 (3SF)	0	EOD	
Note (2): Personal communication FMEO							
3	25	0	Inf. Dilution	E:0.9 (1SF)	0	CRV	22519
4	25	0	Inf. Dilution	E:0.727 (3SF)	0	MEF	22522
Note (4): From the values given for Po+4/Po+3 & Po+3/Po+2							

5	25	0	Inf. Dilution	E:0.363(3SF)	0	CRV	22522
Note (5): From the paper, p. 2870, evidently in error							

Reaction No. 73766							
$\text{Po(s)} = \text{Po(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:109.410(6SF) kJ	4	CRV	3006
2	25	0	Inf. Dilution	dG:109.4(4SF) kJ	4	CRV	17631
3	25	0	Inf. Dilution	dH:147.0(4SF) kJ	4	CRV	17631

Reaction No. 73767							
$2\langle\text{Po(s)}\rangle = \text{Po}_2\text{(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:98.257(5SF) kJ	4	CRV	3006
2	25	0	Inf. Dilution	dG:98.3(3SF) kJ	4	CRV	17631
3	25	0	Inf. Dilution	dH:145.5(4SF) kJ	4	CRV	17631

Reaction No. 73768							
$\text{Po(s)} + 2\langle\text{H+1}\rangle + 2\langle\text{e-1}\rangle = \text{H+1(2)}_{\text{Po-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.07(3SF)	4	CRV	17631
2	25	0	Inf. Dilution	E:-1.0(2SF)	2	EOD	22537

Reaction No. 73769							
$\text{H}_2\text{(g)} + \text{Po(s)} = \text{H+1(2)}_{\text{Po-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-35.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:206.5(4SF) kJ	5	CRV	17631
3	25	0	Inf. Dilution	dG:206.5(4SF) kJ	5	CRV	31246

Reaction No. 73770							
$\text{H+1(2)}_{\text{Po-2}} = \text{H+1}_{\text{Po-2}} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.55(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.49(2SF)	5	CRV	17631
3	25	0	Inf. Dilution	lgK:-0.49(2SF)	5	CRV	31246
Note (3): Table 2, p. 28							

Reaction No. 73771							
$0.5\text{H}_2(\text{g}) + \text{Po}(\text{s}) + \text{e}^{-1} = \text{H}+1_{\text{Po}-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -36.35 (4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: 209.3 (4\text{SF}) \text{ kJ}$	5	CRV	17631
3	25	0	Inf. Dilution	$\text{dG}: 209.3 (4\text{SF}) \text{ kJ}$	5	CRV	31246

Reaction No. 73772							
$\text{H}+1_{\text{Po}-2} = \text{Po}-2 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -11.33 (4\text{SF})$	5	CRV	17631
2	25	0	Inf. Dilution	$\lg K: -11.33 (4\text{SF})$	5	CRV	31246
Note (2): Table 2, p. 28							

Reaction No. 73773							
$\text{Po}(\text{s}) + 2\text{e}^{-1} = \text{Po}-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -48.0 (3\text{SF})$	1	EOD	
Note (1): Personal communication FMEO							
2	25	0	Inf. Dilution	$\text{dG}: 274.0 (4\text{SF}) \text{ kJ}$	4	CRV	17631
3	25	0	Inf. Dilution	$\text{dG}: 274.0 (4\text{SF}) \text{ kJ}$	5	CRV	31246
Note (3): Table 2, p. 28							
4	25	0	Inf. Dilution	$\text{E}: -1.4 (2\text{SF})$	3	CRV	3006
Note (4): Sign changed							
5	25	0	Inf. Dilution	$\text{E}: -1.42 (3\text{SF})$	4	CRV	17631
6	25	0	Inf. Dilution	$\text{E}: -1.4 (2\text{SF})$	3	EOD	22537
7	25	0	Inf. Dilution	$\text{E}: -1.4 (2\text{SF})$	5	CRV	22551
Note (7): Sign changed!							

Reaction No. 73774							
$\text{Po}+2 + 2\text{e}^{-1} = \text{Po}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -71. (2\text{SF}) \text{ kJ}$	0	CRV	3006
Note (1): Sign changed (gives $E_0 = 0.37$)							
2	25	0	Inf. Dilution	$\text{dG}: -124.1 (4\text{SF}) \text{ kJ}$	4	CRV	17631
3	25	0	Inf. Dilution	$\text{dG}: -124.1 (4\text{SF}) \text{ kJ}$	5	CRV	31246

Note (3): Table 2, p. 28							
4	25	0	Inf. Dilution	E:0.368(3SF)	0	CRV	3006
5	25	0	Inf. Dilution	E:0.643(3SF)	4	CRV	17631
6	25	0	Inf. Dilution	E:0.68(0.01SD)	3	MEF	22521
7	25	0	Inf. Dilution	E:0.65(2SF)	3	EOD	22537
8	25	0	Inf. Dilution	E:0.37(2SF)	0	CRV	22551
Note (8): This E0 refers to a measurement in (strongly complexing) HCl							
9	25	0	Inf. Dilution	E:0.643(3SF)	5	CRV	31246

Reaction No. 73775							
$\text{PoO}_2(\text{s}) + 2\text{e}^- + 4\text{H}^+ = \text{Po}^{2+} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.095(4SF)	0	CRV	3006
2	25	0	Inf. Dilution	E:0.819(3SF)	4	CRV	17631
3	25	0	Inf. Dilution	E:0.798(3SF)	3	EOD	22537
4	25	0	Inf. Dilution	E:1.1(2SF)	0	CRV	22551

Reaction No. 73776							
$\text{PoO}^{2+} + 2\text{H}^+ + 2\text{e}^- = \text{Po}^{2+} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.935(3SF)	0	CRV	17631

Reaction No. 73777							
$\text{PoO}^{2+} + \text{H}_2\text{O} = \text{PoO}^{+2}_{\text{OH}-1} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	(H,Na)ClO ₄	lgK:-1.06(3SF)	0	MDS	22552
2	25	0	Inf. Dilution	lgK:-1.2(2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-0.85(2SF)	0	CRV	17631

Reaction No. 73778							
$0.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) - 2\text{e}^- = \text{PoO}^{2+}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:67.4(3SF) kJ	0	CRV	17631
Note (2): Rejected by 17Hum_31246							

Reaction No. 73779							
$\text{PoO}+2_{\text{OH}-1} + \text{H}_2\text{O} = \text{H}+1(2)_{\text{PoO}3-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.67(3\text{SF})$	4	CRV	17631

Reaction No. 73780							
$0.5<\text{H}_2(\text{g})> + \text{O}_2(\text{g}) + \text{Po}(\text{s}) - \text{e}-1 = \text{PoO}+2_{\text{OH}-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -164.9(4\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73781							
$\text{PoO}_2(\text{s}) + \text{H}_2\text{O} = \text{H}+1(2)_{\text{PoO}3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -6.45(3\text{SF})$	2	CRV	17631
Note (1): Deprecated by 17Hum_31246							

Reaction No. 73782							
$\text{H}_2(\text{g}) + 1.5<\text{O}_2(\text{g})> + \text{Po}(\text{s}) = \text{H}+1(2)_{\text{PoO}3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -392.6(4\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73783							
$\text{PoO}_2(\text{s}) + \text{OH}-1 = \text{H}+1_{\text{PoO}3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -4.98(3\text{SF})$	4	CRV	17631

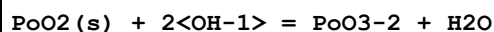
Reaction No. 73784							
$0.5<\text{H}_2(\text{g})> + 1.5<\text{O}_2(\text{g})> + \text{Po}(\text{s}) + \text{e}-1 = \text{H}+1_{\text{PoO}3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -321.1(4\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73785							
$\text{H}+1(2)_{\text{PoO}3-2} = \text{H}+1_{\text{PoO}3-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -12.52(4\text{SF})$	4	CRV	17631

Reaction No. 73786							
$\text{H}+1_{\text{PoO}3-2} = \text{PoO}3-2 + \text{H}+1$							

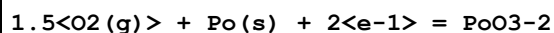
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.45(4SF)	4	CRV	17631

Reaction No. 73787



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.43(3SF)	4	CRV	17631

Reaction No. 73788



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-423.(3SF)kJ	0	CRV	3006
2	25	0	Inf. Dilution	dG:-244.3(4SF)kJ	4	CRV	17631

Reaction No. 73789



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.3(2SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.494(3SF)	0	CRV	3006
3	25	0	Inf. Dilution	E:-0.5(1SF)	0	CRV	14620

Note (3): Zero wgt for inter-reaction consistency

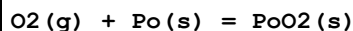
4	25	0	Inf. Dilution	E:-0.031(2SF)	3	CRV	17631
5	25	0	Inf. Dilution	E:0.748(3SF)	0	CRV	22537
6	25	0	Inf. Dilution	E:-0.5(1SF)	0	CRV	22551

Reaction No. 73790



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.43(4SF)	4	EOD	31246
2	25	0	Inf. Dilution	E:0.731(3SF)	4	CRV	17631
3	25	0	Inf. Dilution	E:0.724(3SF)	0	EOD	22537
4	25	0	Inf. Dilution	E:0.73(2SF)	5	CRV	22551

Reaction No. 73791



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-912.(3SF)kJ	0	CRV	3006

Note (1): Prob. just dyslexia							
2	25	0	Inf. Dilution	dG:-192.2(4SF)kJ	4	CRV	17631
3	25	0	Inf. Dilution	dG:-192.1(4SF)kJ	5	CRV	31246
Note (3): p. 14 & Table 2, p. 28							
4	25	0	Inf. Dilution	dH:-251.(3SF)kJ	4	MTD	3006
5	25	0	Inf. Dilution	dH:-251.(3SF)kJ	4	CRV	17631

Reaction No. 73792							
$\text{PoO3(s)} + 2\text{e-1} + 2\text{H+1} = \text{PoO2(s)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.509(4SF)	3	CRV	3006
2	25	0	Inf. Dilution	E:1.509(4SF)	4	CRV	17631
3	25	0	Inf. Dilution	E:1.524(4SF)	3	EOD	22537
4	25	0	Inf. Dilution	E:1.51(3SF)	5	CRV	22551

Reaction No. 73793							
$1.5\text{O2(g)} + \text{Po(s)} = \text{PoO3(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-138.(3SF)kJ	4	CRV	3006
2	25	0	Inf. Dilution	dG:-138.2(4SF)kJ	4	CRV	17631

Reaction No. 73794							
$\text{PoO3(s)} + 2\text{e-1} = \text{PoO3-2}$							
Note: Caution advised - see [17Hum_31246, p. 26] (but noted "result in excellent agreement with literature")							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.477(4SF)	0	CRV	3006
2	25	0	Inf. Dilution	E:0.550(3SF)	2	CRV	17631
3	25	0	Inf. Dilution	E:1.474(4SF)	0	CRV	22537
4	25	0	Inf. Dilution	E:1.48(3SF)	0	CRV	22551

Reaction No. 73795							
$\text{Po+4}_{\text{Cl-1(6)}} + 2\text{e-1} = \text{Po+2}_{\text{Cl-1(4)}} + 2\text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.775(3SF)	1	CRV	17631
2	25	1	HCl	E:0.717(3SF)	0	EOD	31246
Note (2): Reactive Cl-1							

3	25	1.5	HCl	E:0.702 (3SF)	0	EOD	31246
Note (3): Reactive Cl-1							

Reaction No. 73796							
$3\text{Cl}_2(\text{g}) + \text{Po}(\text{s}) + 2\text{e}^- = \text{Po+4_Cl-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-577.(3SF)kJ	0	CRV	3006
2	25	0	Inf. Dilution	dG:-554.0(4SF)kJ	2	CRV	17631
3	25	0	Inf. Dilution	dG:-555.3(4SF)kJ	5	CRV	31246
Note (3): Table 2, p. 28							

Reaction No. 73797							
$\text{PoO}_2 + 2\text{H}^+ + 6\text{Cl}^- = \text{Po+4_Cl-1(6)} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.7(3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:12.7(3SF)	5	CRV	31246

Reaction No. 73798							
$\text{Po+4_Cl-1(6)} + 4\text{e}^- = \text{Po}(\text{s}) + 6\text{Cl}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.601(3SF)	4	CRV	17631

Reaction No. 73799							
$\text{Po+2_Cl-1(4)} + 2\text{e}^- = \text{Po}(\text{s}) + 4\text{Cl}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.420(3SF)	4	CRV	17631

Reaction No. 73800							
$2\text{Cl}_2(\text{g}) + \text{Po}(\text{s}) + 2\text{e}^- = \text{Po+2_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-443.0(4SF)kJ	5	CRV	17631
2	25	0	Inf. Dilution	dG:-443.9(4SF)kJ	5	CRV	31246
Note (2): Table 2, p. 28							

Reaction No. 73801							
$4\text{Cl}^- + \text{Po+2} = \text{Po+2_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:7.55 (3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:7.55 (3SF)	5	CRV	31246
Note (2): Table 2, p. 28							
3	25	0	Unknown	lgK:7.55 (3SF)	3	EIE	31585
Note (3): Table Supp. 3							

Reaction No. 73802							
$3\text{Cl-1} + \text{Po+2} = \text{Po+2_Cl-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.02 (3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:7.02 (3SF)	5	CRV	31246
Note (2): Table 2, p. 28							
3	25	0	Unknown	lgK:7.39 (3SF)	3	EIE	31585
Note (3): Table Supp. 3							

Reaction No. 73803							
$1.5\text{Cl}_2(\text{g}) + \text{Po}(\text{s}) + \text{e-1} = \text{Po+2_Cl-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-309.0 (4SF) kJ	4	CRV	17631

Reaction No. 73804							
$2\text{Cl-1} + \text{Po+2} = \text{Po+2_Cl-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.13 (3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:6.13 (3SF)	5	CRV	31246
Note (2): Table 2, p. 28							
3	25	0	Unknown	lgK:6.85 (3SF)	3	EIE	31585
Note (3): Table Supp. 3							

Reaction No. 73805							
$\text{Cl}_2(\text{g}) + \text{Po}(\text{s}) = \text{Po+2_Cl-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-172.9 (4SF) kJ	4	CRV	17631

Reaction No. 73806							
$\text{Cl-1} + \text{Po+2} = \text{Po+2_Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:3.75(3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:3.75(3SF)	5	CRV	31246
Note (2): Table 2, p. 28							
3	25	0	Unknown	lgK:4.26(3SF)	3	EIE	31585
Note (3): Table Supp. 3							

Reaction No. 73807							
$0.5\text{Cl}_2(\text{g}) + \text{Po}(\text{s}) - \text{e}^- = \text{Po} + 2\text{Cl}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-28.3(3SF) kJ	4	CRV	17631

Reaction No. 73808							
$\text{Po} + 4\text{Cl}^- (4)_{\text{OH}^-} + \text{H}^+ + 2\text{Cl}^- = \text{Po} + 4\text{Cl}^- (6) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.16(3SF)	2	CRV	17631
2	25	0	Inf. Dilution	lgK:2.16(3SF)	5	CRV	31246
Note (2): Table 2, p. 28							

Reaction No. 73809							
$2\text{Cl}_2(\text{g}) + 0.5\text{H}_2(\text{g}) + 0.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) + \text{e}^- = \text{Po} + 4\text{Cl}^- (4)_{\text{OH}^-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:90.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-516.9(4SF) kJ	0	CRV	17631

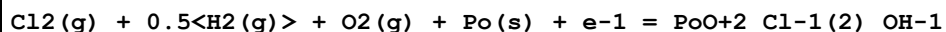
Reaction No. 73810							
$\text{Po} + 2\text{Cl}^- (4) + 2\text{H}_2\text{O} = \text{H}^+ (2)_{\text{PoO}_3} + 2\text{H}^+ + 4\text{Cl}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.9(2SF)	4	CRV	17631

Reaction No. 73811							
$2\text{Cl}_2(\text{g}) + 0.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) + 2\text{e}^- = \text{Po} + 2\text{Cl}^- (4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-493.1(4SF) kJ	4	CRV	17631

Reaction No. 73812							
$\text{Po} + 2\text{Cl}^- (2)_{\text{OH}^-} + \text{H}_2\text{O} = \text{H}^+ (2)_{\text{PoO}_3} + \text{H}^+ + 2\text{Cl}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

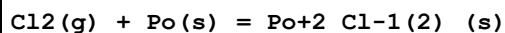
1	25	0	Inf. Dilution	lgK:-4.8(2SF)	4	CRV	17631
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Reaction No. 73813



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-444.8(4SF)kJ	4	CRV	17631

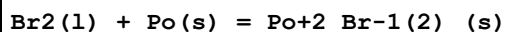
Reaction No. 73814



Note: Reaction deliberately excluded from TDB 2020 [17Hum_31246, p. 8] (Data estimated by Brown (2001) for highly soluble solids)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-130.9(4SF)kJ	5	CRV	17631
2	25	0	Inf. Dilution	dG:-130.9(4SF)kJ	5	CRV	31246
3	25	0	Inf. Dilution	dH:-177.4(4SF)kJ	4	CRV	17631

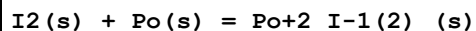
Reaction No. 73815



Note: Reaction deliberately excluded from TDB 2020 [17Hum_31246, p. 8] (Data estimated by Brown (2001) for highly soluble solids)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-50.4(3SF)kJ	5	CRV	17631
2	25	0	Inf. Dilution	dG:-50.4(3SF)kJ	5	CRV	31246
3	25	0	Inf. Dilution	dH:-68.3(3SF)kJ	4	CRV	17631

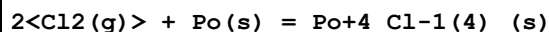
Reaction No. 73816



Note: Reaction deliberately excluded from TDB 2020 [17Hum_31246, p. 8] (Species HYPOTHESISED by Brown (2001))

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:56.7(3SF)kJ	4	CRV	17631
2	25	0	Inf. Dilution	dH:76.3(3SF)kJ	4	CRV	17631

Reaction No. 73817



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-242.7(4SF)kJ	4	CRV	17631
2	25	0	Inf. Dilution	dH:-332.7(4SF)kJ	4	CRV	17631

Reaction No. 73818							
$\text{Po+4_Cl-1(4)_(s)} + 2\text{<Cl-1>} = \text{Po+4_Cl-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.63(3SF)	4	CRV	17631

Reaction No. 73819							
$\text{Po+4_Cl-1(4)_(s)} + \text{H}_2\text{O} = \text{PoO+2_Cl-1(4)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.31(3SF)	4	CRV	17631

Reaction No. 73820							
$2\text{<Br2(1)>} + \text{Po(s)} = \text{Po+4_Br-1(4)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-153.4(4SF) kJ	4	CRV	17631
2	25	0	Inf. Dilution	dH:-199.4(4SF) kJ	4	CRV	17631

Reaction No. 73821							
$2\text{<I2(s)>} + \text{Po(s)} = \text{Po+4_I-1(4)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-49.4(3SF) kJ	4	CRV	17631
2	25	0	Inf. Dilution	dH:-53.1(3SF) kJ	4	CRV	17631

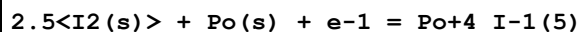
Reaction No. 73822							
$\text{Po+4_I-1(4)_(s)} + 2\text{<I-1>} = \text{Po+4_I-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.02:0.5	(H,Li)I	lgK:-2.23(3SF)	3	MSL	22541
2	25	0	Inf. Dilution	lgK:-2.23(3SF)	4	CRV	17631

Reaction No. 73823							
$3\text{<I2(s)>} + \text{Po(s)} + 2\text{<e-1>} = \text{Po+4_I-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-139.9(4SF) kJ	4	CRV	17631

Reaction No. 73824							
$\text{Po+4_I-1(4)_(s)} + \text{I-1} = \text{Po+4_I-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.02:0.5	(H,Li)I	lgK:-4.17(3SF)	3	MSL	22541

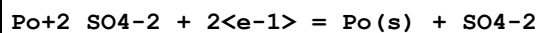
2	25	0	Inf. Dilution	lgK:-4.17(3SF)	4	CRV	17631
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Reaction No. 73825



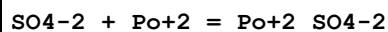
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-77.2(3SF)kJ	4	CRV	17631

Reaction No. 73826



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:20.62(4SF)	5	CRV	31246
3	25	0	Inf. Dilution	E:0.61(2SF)	5	CRV	17631
4	25	0	Inf. Dilution	E:0.61(2SF)	5	CRV	31246

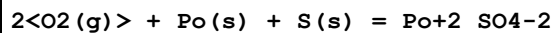
Reaction No. 73827



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.12(3SF)	5	CRV	17631
2	25	0	Inf. Dilution	lgK:1.12(3SF)	5	CRV	31246

Note (2): Table 2, p. 28

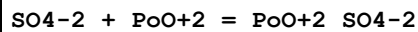
Reaction No. 73828



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:110.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-626.2(4SF)kJ	4	CRV	17631
3	25	0	Inf. Dilution	dG:-626.3(4SF)kJ	5	CRV	31246

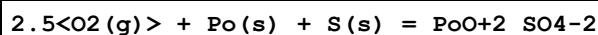
Note (3): Table 2, p. 28

Reaction No. 73829



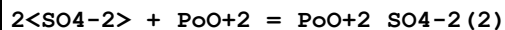
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.56(3SF)	4	CRV	17631

Reaction No. 73830



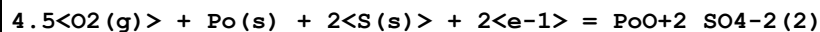
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:121.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-691.1(4SF)kJ	0	CRV	17631

Reaction No. 73831



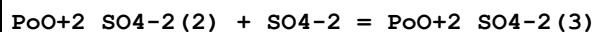
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.50(3SF)	4	CRV	17631

Reaction No. 73832



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:254.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1446.1(5SF)kJ	0	CRV	17631

Reaction No. 73833

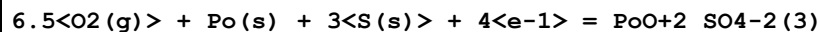


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.4(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.22(3SF)	0	CRV	17631

Note (2): rejected by [17Hum_31246, p. 19]

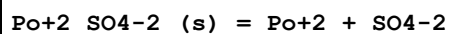
3	25	0	Inf. Dilution	lgK:2.2(2SF)	4	CRV	31246
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Reaction No. 73834



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:385.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2197.0(5SF)kJ	0	CRV	17631

Reaction No. 73835



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.89(3SF)	3	CRV	17631
2	25	0	Inf. Dilution	lgK:-9.97(3SF)	0	CRV	31246

Note (2): Selected by Hummel (2002) but now rejected

3	25	0	Inf. Dilution	lgK:-8.91(3SF)	5	CRV	31246
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Note (3): Table 2, p. 28

Reaction No. 73836							
$2\text{O}_2(\text{g}) + \text{Po}(\text{s}) + \text{S}(\text{s}) = \text{Po}_2\text{S}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 117.8(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -670.6(4\text{SF}) \text{ kJ}$	0	CRV	17631

Reaction No. 73837							
$\text{PoO} \cdot \text{PoO} \cdot \text{OSO}_4(\text{s}) + \text{H}+1(2)\text{SO}_4-2 = 2\text{PoO}_2\text{SO}_4-2 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -10.8(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -9.48(3\text{SF})$	0	CRV	17631
Note (2): Zero wgt for inter-rxn consistency; select rxn ex H+1(2)SO4-2							

Reaction No. 73838							
$3.5\text{O}_2(\text{g}) + 2\text{Po}(\text{s}) + \text{S}(\text{s}) = \text{PoO} \cdot \text{PoO} \cdot \text{OSO}_4(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -937.3(4\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73839							
$\text{PoO} \cdot \text{PoO} \cdot \text{OSO}_4(\text{s}) + \text{H}_2\text{O} = 2\text{PoO}_2 + 2\text{OH}-1 + \text{SO}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -43.9(3\text{SF})$	4	CRV	17631

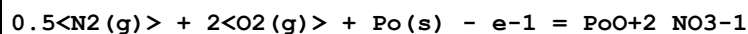
Reaction No. 73840							
$\text{Po}_4\text{SO}_4-2(2)\text{H}_2\text{O}(\text{s}) + \text{SO}_4-2 = \text{PoO}_2\text{SO}_4-2(3) + 2\text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -3.5(2\text{SF})$	4	CRV	17631

Reaction No. 73841							
$\text{H}_2(\text{g}) + 4.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) + 2\text{S}(\text{s}) = \text{Po}_4\text{SO}_4-2(2)\text{H}_2\text{O}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 259.3(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -1473.1(5\text{SF}) \text{ kJ}$	0	CRV	17631

Reaction No. 73842							
$\text{NO}_3-1 + \text{PoO}_2 = \text{PoO}_2\text{NO}_3-1$							
Note: Data in doubt - see [17Hum_31246, p. 23]							

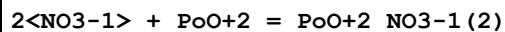
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.29(3SF)	1	CRV	17631

Reaction No. 73843



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-50.5(3SF) kJ	0	CRV	17631

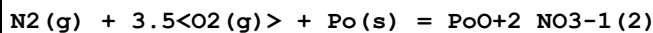
Reaction No. 73844



Note: Data in doubt - see [17Hum_31246, p. 23]

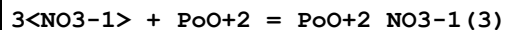
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.31(3SF)	1	CRV	17631

Reaction No. 73845



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-166.8(4SF) kJ	0	CRV	17631

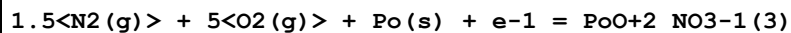
Reaction No. 73846



Note: Data in doubt - see [17Hum_31246, p. 23]

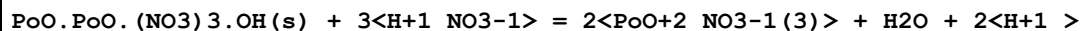
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.33(3SF)	1	CRV	17631

Reaction No. 73847



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-277.4(4SF) kJ	0	CRV	17631

Reaction No. 73848



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.5(2SF)	4	CRV	17631

Reaction No. 73849							
$0.5\text{H}_2(\text{g}) + 1.5\text{N}_2(\text{g}) + 6\text{O}_2(\text{g}) + 2\text{Po}(\text{s}) = \text{PoO} \cdot \text{PoO} \cdot (\text{NO}_3)_3 \cdot \text{OH}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG: -521.0 (4SF) kJ	4	CRV	17631

Reaction No. 73850							
$\text{PoO} \cdot \text{PoO} \cdot (\text{NO}_3)_3 \cdot \text{OH}(\text{s}) = 2\text{PoO} + 2 + \text{OH}^{-1} + 3\text{NO}_3^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: -28.9 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK: -33.9 (3SF)	0	CRV	17631
Note (2): Zero wgt for inter-rxn consistency; other rxn gives better solubility							

Reaction No. 73851							
$\text{SeO}_4^{-2} + \text{PoO} + 2 = \text{PoO} + 2_{\text{SeO}_4^{-2}}$							
Note: Data in doubt - see [17Hum_31246, p. 22]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 2.1 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK: 2.43 (3SF)	0	CRV	17631
Note (2): p. 80							

Reaction No. 73852							
$2\text{SeO}_4^{-2} + \text{PoO} + 2 = \text{PoO} + 2_{\text{SeO}_4^{-2}}(2)$							
Note: Data in doubt - see [17Hum_31246, p. 22]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 3.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK: 3.78 (3SF)	0	CRV	17631
Note (2): p. 80							

Reaction No. 73853							
$2.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) + \text{Se}(\text{s}) = \text{PoO} + 2_{\text{SeO}_4^{-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG: -387.4 (4SF) kJ	4	CRV	17631
Note (1): p. 80							

Reaction No. 73854							
$4.5\text{O}_2(\text{g}) + \text{Po}(\text{s}) + 2\text{Se}(\text{s}) + 2\text{e}^{-1} = \text{PoO} + 2_{\text{SeO}_4^{-2}}(2)$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-836.0(4SF)kJ	4	CRV	17631
Note (1): p. 80							

Reaction No. 73855							
$\text{PoO.PoO.OSeO}_4(\text{s}) + \text{H}+1(2)\text{SeO}_4-2 = 2\text{PoO}+2\text{SeO}_4-2 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.97(3SF)	0	CRV	17631
Note (2): p. 81							

Reaction No. 73856							
$\text{PoO.PoO.OSeO}_4(\text{s}) + 3\text{H}+1(2)\text{SeO}_4-2 = 2\text{PoO}+2\text{SeO}_4-2(2) + \text{H}_2\text{O} + 4\text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.8(2SF)	0	CRV	17631
Note (2): p. 82 (the sign on the p/c is unclear; neither way works)							

Reaction No. 73857							
$\text{PoO.PoO.OSeO}_4(\text{s}) + \text{H}_2\text{O} = 2\text{PoO}+2 + 2\text{OH}-1 + \text{SeO}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-43.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-43.1(3SF)	0	CRV	17631
Note (2): p. 82							

Reaction No. 73858							
$3.5\text{O}_2(\text{g}) + 2\text{Po}(\text{s}) + \text{Se}(\text{s}) = \text{PoO.PoO.OSeO}_4(\text{s})$							
Note: Data located on p. 82							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-628.0(4SF)kJ	4	CRV	17631
2	25	0	Inf. Dilution	dG:-631.0(4SF)kJ	4	CRV	17631
3	25	0	Inf. Dilution	dG:-629.5(4SF)kJ	4	CRV	17631

Reaction No. 73859							
$\text{PoO}+2 + 6\text{CN}-1 + 2\text{H}+1 = \text{Po}+4\text{CN}-1(6) + \text{H}_2\text{O}$							
Note: Data in doubt - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:53.8(3SF)	1	EES	
2	25	0	Inf. Dilution	lgK:49.8(3SF)	0	CRV	17631

Reaction No. 73860							
$6\text{C(s)} + 3\text{N}_2\text{(g)} + \text{Po(s)} + 2\text{e}^- = \text{Po+4}_{\text{CN-1}}\text{(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:1013.8(5SF)kJ	4	CRV	17631

Reaction No. 73861							
$2\text{PoO+2}_{\text{CN-1}}\text{(2)}_{\text{(s)}} + 2\text{CN-1} = \text{PoO}_2\text{(s)} + \text{Po+4}_{\text{CN-1}}\text{(6)}$							
Note: Data in doubt - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.9(2SF)	1	CRV	17631

Reaction No. 73862							
$2\text{PoO+2}_{\text{CN-1}}\text{(2)}_{\text{(s)}} + 2\text{CN-1} + \text{OH-1} = \text{H+1}_{\text{PoO3-2}} + \text{Po+4}_{\text{CN-1}}\text{(6)}$							
Note: Data in doubt - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.86(3SF)	1	CRV	17631

Reaction No. 73863							
$2\text{C(s)} + \text{N}_2\text{(g)} + 0.5\text{O}_2\text{(g)} + \text{Po(s)} = \text{PoO+2}_{\text{CN-1}}\text{(2)}_{\text{(s)}}$							
Note: Data in doubt - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:234.1(4SF)kJ	1	CRV	17631

Reaction No. 73864							
$\text{Po+2}_{\text{S-2}}\text{(s)} = \text{Po+2} + \text{S-2}$							
Note: The species S-2 does not exist in aqueous solution See reaction: $\text{H+1} + \text{S-2} = \text{H+1}_{\text{S-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.3(3SF)	0	CRV	14620
Note (1): p. 783, no correction for I							
2	25	0	Inf. Dilution	lgK:-27.26(4SF)	0	CRV	17631
3	25	0	Inf. Dilution	lgK:-28.26(4SF)	0	EOD	22537
Note (3): No correction for I it seems							
4	25	2	HCl	lgK:-28.27(4SF)	0	MSL	22543

Reaction No. 73865							
$\text{Po(s)} + \text{S(s)} = \text{Po}_{2}\text{S}_{-2}\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -9.9 (2\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: 218. (3\text{SF}) \text{ kJ}$	0	CRV	3006
3	25	0	Inf. Dilution	$\text{dG}: 61.0 (3\text{SF}) \text{ kJ}$	0	CRV	17631

Reaction No. 73866							
$\text{Ag}_{2}\text{PoO}_{3}(\text{s}) = 2\text{Ag}^{+1} + \text{PoO}_{3-2}$							
Note: Caution advised - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -24.40 (4\text{SF})$	2	CRV	17631

Reaction No. 73867							
$2\text{Ag}(\text{s}) + 1.5\text{O}_{2}(\text{g}) + \text{Po}(\text{s}) = \text{Ag}_{2}\text{PoO}_{3}(\text{s})$							
Note: Caution advised - see [17Hum_31246, p. 26]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{dG}: -228.2 (4\text{SF}) \text{ kJ}$	2	CRV	17631

Reaction No. 73868							
$\text{PbPo}(\text{s}) = \text{Pb}^{+2} + \text{Po}^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -55.2 (3\text{SF})$	5	CRV	17631
2	25	0	Inf. Dilution	$\lg K: -55.2 (3\text{SF})$	5	CRV	31246

Reaction No. 73869							
$\text{HgPo}(\text{s}) = \text{Hg}^{+2} + \text{Po}^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -76.8 (3\text{SF})$	5	CRV	17631
2	25	0	Inf. Dilution	$\lg K: -76.8 (3\text{SF})$	5	CRV	31246

Reaction No. 73870							
$\text{ZnPo}(\text{s}) = \text{Zn}^{+2} + \text{Po}^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -37.7 (3\text{SF})$	5	CRV	17631
2	25	0	Inf. Dilution	$\lg K: -37.7 (3\text{SF})$	5	CRV	31246

Reaction No. 73871							
$\text{NiPo(s)} = \text{Ni}^{+2} + \text{Po}^{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -46.1(3\text{SF})$	5	CRV	17631
2	25	0	Inf. Dilution	$\lg K: -46.1(3\text{SF})$	5	CRV	31246

Reaction No. 73872							
$\text{Pb(s)} + \text{Po(s)} = \text{PbPo(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\Delta G: -65.3(3\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73873							
$\text{Hg(l)} + \text{Po(s)} = \text{HgPo(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\Delta G: -0.8(1\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73874							
$\text{Po(s)} + \text{Zn(s)} = \text{ZnPo(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\Delta G: -88.7(3\text{SF}) \text{ kJ}$	4	CRV	17631

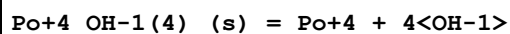
Reaction No. 73875							
$\text{Ni(s)} + \text{Po(s)} = \text{NiPo(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\Delta G: -34.0(3\text{SF}) \text{ kJ}$	4	CRV	17631

Reaction No. 73881							
$\text{Po}^{+2}_{\text{S-2}}(\text{s}) + \text{H}^{+1} = \text{Po}^{+2} + \text{H}^{+1}_{\text{S-2}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -14.2(3\text{SF})$	1	ELC	
Note (1): Also accords better with $\lg K$ for analogous rxn with Pb^{+2}							
2	25	0	Inf. Dilution	$\lg K: 23.6(3\text{SF})$	0	CRV	5081
Note (2): Zero wgt for inter-reaction consistency							

Reaction No. 73981							
$\text{PoO}^{+2}_{\text{OH-1}(2)}(\text{s}) + 2\langle\text{OH-1}\rangle = \text{PoO}^{+2}_{\text{S-2}} + 2\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-4.086(4SF)	3	CRV	14620
2	25	0	K+1 salt	lgK:-4.086(0.02SD)	3	MSL	22534

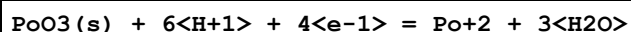
Reaction No. 73982



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-37.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-38.22(4SF)	0	MSL	22536

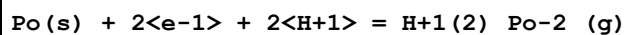
Note (2): Refers to Po+4 and should not

Reaction No. 73983



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.16(3SF)	5	CRV	22537
2	25	0	Inf. Dilution	E:1.3(2SF)	0	CRV	22551

Reaction No. 73984



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.(1SF)	3	CRV	3006
2	25	0	Inf. Dilution	E:1.(1SF)	3	CRV	22551

Reaction No. 73985

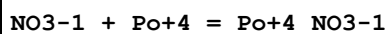


No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.99(2SF)	1	ELC	

Note (1): Note lgK = 100 eqv. E = 1V

2	25	0	Inf. Dilution	E:0.16(2SF)	0	CRV	22551
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Reaction No. 73992



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:0.56(2SF)	0	MIX	22552
2	25	0	Inf. Dilution	lgK:2.0(0.45SD)	3	CRV	31246

Note (2): Table 2, p. 28

Reaction No. 73993

2<NO3-1> + Po+4 = Po+4_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:1.15(3SF)	0	MIX	22552
2	25	0	Inf. Dilution	lgK:3.7(0.2SD)	3	CRV	31246
Note (2): Table 2, p. 28							

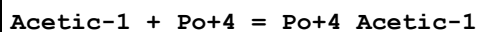
Reaction No. 73994							
3<NO3-1> + Po+4 = Po+4_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:1.30(3SF)	0	MIX	22552
2	25	0	Inf. Dilution	lgK:4.4(0.25SD)	3	CRV	31246
Note (2): Table 2, p. 28							

Reaction No. 73995							
SO4-2 + Po+4 = Po+4_SO4-2							
Note: See comparisons with other M+4 cations [17Hum_31246, p. 18] which support confidence in this result							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:1.5(2SF)	0	MIX	22552
2	25	0	Inf. Dilution	lgK:5.2(0.3SD)	4	CRV	31246
Note (2): see remark above							
3	25	0	Inf. Dilution	lgK:2.56(3SF)	1	EOD	31246
Note (3): "Shortcomings" in interpretation [17Hum_31246]							
4	25	2	Unknown	lgK:1.46(3SF)	1	EOD	31246
Note (4): "Shortcomings" in interpretation [17Hum_31246]							

Reaction No. 73996							
2<SO4-2> + Po+4 = Po+4_SO4-2(2)							
Note: See comparisons with other M+4 cations [17Hum_31246, p. 18] which support confidence in this result							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:3.38(3SF)	0	MIX	22552
2	25	0	Inf. Dilution	lgK:9.1(0.35SD)	3	CRV	31246
Note (2): see remark above							
3	25	0	Inf. Dilution	lgK:4.50(3SF)	0	EOD	31246
Note (3): "Shortcomings" in interpretation [17Hum_31246]							
4	25	2	Unknown	lgK:3.40(3SF)	0	EOD	31246

Note (4): "Shortcomings" in interpretation [17Hum_31246]

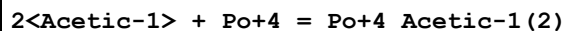
Reaction No. 73997



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:2.51(3SF)	1	MIX	22552

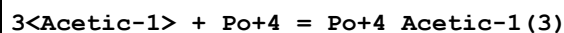
Note (1): Prob. should be wrt PoO+2

Reaction No. 73998



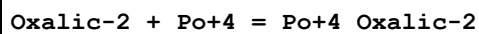
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:4.85(3SF)	3	MIX	22552

Reaction No. 73999



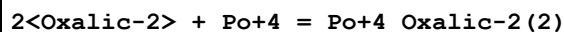
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:7.18(3SF)	3	MIX	22552

Reaction No. 74000



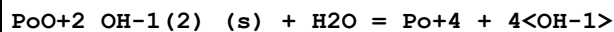
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:5.23(3SF)	3	MIX	22552
2	25	0	Inf. Dilution	lgK:5.2(2SF)	1	ELC	

Reaction No. 74001



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:9.74(3SF)	3	MIX	22552
2	25	0	Inf. Dilution	lgK:9.7(2SF)	1	ESC	

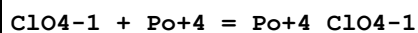
Reaction No. 74002



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.0(3SF)	0	MEF	22555

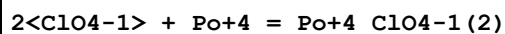
Note (1): Refers to Po+4 but should not

Reaction No. 74003



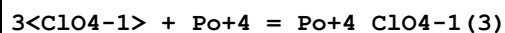
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1:4	Unknown	lgK:-0.89 (2SF)	3	MDS	22556

Reaction No. 74004



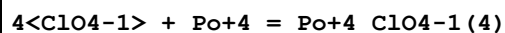
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1:4	Unknown	lgK:-1.48 (3SF)	3	MDS	22556

Reaction No. 74005



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1:4	Unknown	lgK:-2.05 (3SF)	3	MDS	22556

Reaction No. 74006



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1:4	Unknown	lgK:-2.78 (3SF)	3	MDS	22556

Reaction No. 74007



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.16 (2SF)	0	CRV	14620
3	25	0	Inf. Dilution	E:0.857 (3SF)	0	CRV	22537

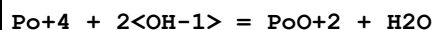
Note (3): Bad inter-reaction consistency

Reaction No. 74008



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.847 (3SF)	0	CRV	22537

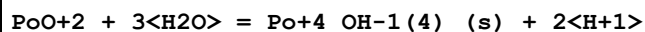
Reaction No. 74009



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:30. (2SF)	0	CDF	

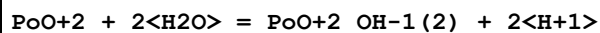
Note (2): Balances hydrolysis of Po+4 to PoO+2 & E0 for Po+4_Cl-1(6) wrt Po(s)

Reaction No. 74010



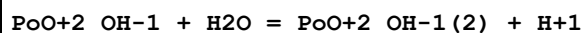
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.22(4SF)	4	EOD	22559

Reaction No. 74011



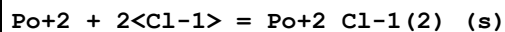
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	(H,Na)ClO4	lgK:-3.26(3SF)	3	MDS	22552
2	25	0	Inf. Dilution	lgK:-3.06(3SF)	4	CRV	17631

Reaction No. 74012



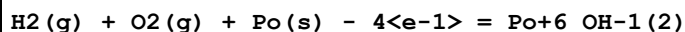
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.21(3SF)	0	CRV	17631

Reaction No. 74013



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.115(4SF)	1	ELC	

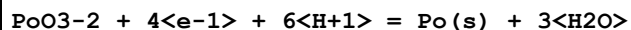
Reaction No. 74016



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-473.(3SF)kJ	0	CRV	3006

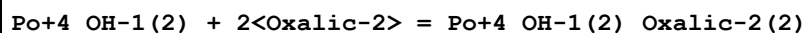
Note (1): No evidence for Po+6

Reaction No. 74017



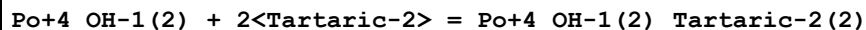
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:81.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.748(3SF)	0	CRV	3006

Reaction No. 74018



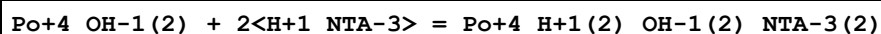
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:7.78 (3SF)	3	MDS	22562
2	25	0.4	NaClO4	lgK:7.48 (3SF)	3	MIX	22562

Reaction No. 74019



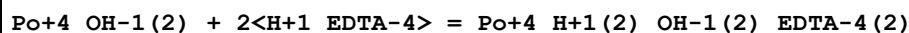
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:7.30 (3SF)	3	MDS	22562
2	25	0.4	NaClO4	lgK:7.90 (3SF)	3	MIX	22562

Reaction No. 74020



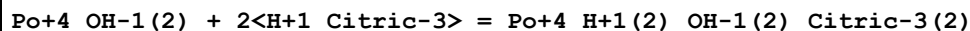
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:8.18 (3SF)	0	MDS	22562
2	25	0.4	NaClO4	lgK:5.78 (3SF)	0	MIX	22562

Reaction No. 74021



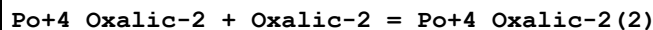
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:8.0 (2SF)	0	MDS	22562
2	25	0.4	NaClO4	lgK:5.95 (3SF)	0	MIX	22562

Reaction No. 74022



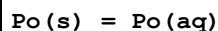
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:8.18 (3SF)	0	MDS	22562
2	25	0.4	NaClO4	lgK:5.78 (3SF)	0	MIX	22562

Reaction No. 75945



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaClO4	lgK:4.51 (3SF)	5	MIX	22552
2	25	0	Inf. Dilution	lgK:4.5 (2SF)	1	ESC	

Reaction No. 80509



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:-7.(1SF)	3	CRV	31246
Note (1): By analogy with Pb; see also Table 2, p. 28							

Reaction No. 80510							
$\text{Po+2_S-2(s)} + 2\text{<H+1>} = \text{Po+2} + \text{H+1(2)_S-2}$							
Note: Considered by 17Hum_31246 but omitted from TDB 2020							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.0(2SF)	0	EES	

Reaction No. 80511							
$\text{PoO2(s)} + 4\text{<H+1>} = \text{Po+4} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.(1SF)	3	CRV	31246
Note (1): lgK "very poorly defined"; see also Table 2, p. 28							

Reaction No. 80512							
$\text{PoO2(s)} + 2\text{<H2O>} + 2\text{<OH-1>} = \text{Po+4_OH-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.43(0.05SD)	4	CRV	31246

Reaction No. 80513							
$\text{PoO2(s)} + 4\text{<H2O>} = \text{Po+4_OH-1(6)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.43(0.05SD)	4	CRV	31246
Note (1): Table 2, p. 28							

Reaction No. 80514							
$\text{PoO2(s)} + 2\text{<H2O>} = \text{Po+4_OH-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.5(2SF)	4	CRV	31246
Note (1): Table 2, p. 28							

Reaction No. 80515							
$\text{PoO2(s)} + \text{H+1} + \text{H2O} = \text{Po+4_OH-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.0(2SF)	4	CRV	31246
Note (1): Table 2, p. 28							

Reaction No. 80516							
$\text{PoO}_2(\text{s}) + 2\text{H}^+ = \text{Po}^{4+} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.5(1\text{SF})$	2	CRV	31246
Note (1): approx; see also Table 2, p. 28							

Reaction No. 80517							
$\text{PoO}_2(\text{s}) + 3\text{H}^+ = \text{Po}^{4+} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 0.5(1\text{SF})$	2	CRV	31246
Note (1): approx; see also Table 2, p. 28							

Reaction No. 80518							
$\text{Po}^{4+} + \text{H}_2\text{O} = \text{Po}^{4+} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.5(1\text{SF})$	2	CRV	31246
Note (1): approx							

Reaction No. 80519							
$\text{Po}^{4+} + \text{H}_2\text{O} = \text{Po}^{4+} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.0(2\text{SF})$	2	CRV	31246
Note (1): approx							

Reaction No. 80520							
$\text{Po}^{4+} + \text{H}_2\text{O} = \text{Po}^{4+} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -2.5(2\text{SF})$	2	CRV	31246
Note (1): approx							

Reaction No. 80521							
$\text{Po}^{4+} + \text{H}_2\text{O} = \text{Po}^{4+} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -4.5(2\text{SF})$	2	CRV	31246
Note (1): approx							

Reaction No. 80522							
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Po+4 + 2<H2O> = Po+4_OH-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.5(2SF)	2	CRV	31246
Note (1): approx							

Reaction No. 80523							
Po+4 + 3<H2O> = Po+4_OH-1(3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.0(2SF)	2	CRV	31246
Note (1): approx							

Reaction No. 80524							
Po+4 + 4<H2O> = Po+4_OH-1(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.5(2SF)	2	CRV	31246
Note (1): approx							

Reaction No. 80649							
PoO+2_Cl-1(4) + 2<H2O> = PoO+2_OH-1(2) + 2<H+1> + 4<Cl-1>							
Note: Uncertainty over state of hydrolysis. Also seems incompatible with solubility data [17Hum_31246, p. 16]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.9(2SF)	2	EOD	31246
2	25	1	(H,Na)Cl	lgK:-8.7(2SF)	2	EOD	31246

Reaction No. 80650							
PoO+2_Cl-1(2)_OH-1 + H2O = PoO+2_OH-1(2) + 2<Cl-1> + H+1							
Note: Uncertainty over state of hydrolysis. Also seems incompatible with solubility data [17Hum_31246, p. 16]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.8(2SF)	2	EOD	31246
2	25	1	(H,Na)Cl	lgK:-4.6(2SF)	2	EOD	31246

Reaction No. 80651							
Po+4 + H+1_SO4-2 = Po+4_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.26(0.3SD)	4	CRV	31246
2	25	2	Unknown	lgK:-0.10(0.11SD)	4	CRV	31246

Reaction No. 80652							
$\text{Po+4} + 2\text{<H+1_SO4-2>} = \text{Po+4_SO4-2(2)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.10(0.35SD)	2	CRV	31246
2	25	2	Unknown	lgK:0.26(0.15SD)	2	CRV	31246

Reaction No. 80653							
$\text{Po+4_SO4-2} + \text{SO4-2} = \text{Po+4_SO4-2(2)}$							
Note: See comparisons with other M+4 cations [17Hum_31246, p. 18] which support confidence in this result							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.9(0.45SD)	5	CRV	31246
Note (1): see remark above							

Reaction No. 80654							
$\text{Po+4_SO4-2(2)_H2O(s)} + \text{H+1_SO4-2} = \text{Po+4_SO4-2(3)} + \text{H+1} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.33(0.02SD)	5	CRV	31246
Note (1): Table 2, p. 28							
2	25	4	H2SO4	lgR:-6.1(2SF)	1	ESI	31246

Reaction No. 80667							
$\text{Po+4_OH-1(2)} + \text{NO3-1} = \text{Po+4_NO3-1_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.3(0.45SD)	3	CRV	31246

Reaction No. 80668							
$\text{Po+4_OH-1(2)} + 2\text{<NO3-1>} = \text{Po+4_NO3-1(2)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.3(0.2SD)	3	CRV	31246

Reaction No. 80669							
$\text{Po+4_OH-1(2)} + 3\text{<NO3-1>} = \text{Po+4_NO3-1(3)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.3(0.25SD)	3	CRV	31246

Reaction No. 80670							
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H2(g) + Po(s) - 4<e-1> = Po+2_H+1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:206.5(4SF)kJ	4	CRV	31246
Note (1): Table 2, p. 28							

Reaction No. 80671							
0.5<H2(g)> + Po(s) - 3<e-1> = Po+2_H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:209.3(4SF)kJ	4	CRV	31246
Note (1): Table 2, p. 28							

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CDF	Definition
CRV	Critical review
EES	Personal estimate
EIE	Interpolation/Extrapolation (not Debye-Huekel)
ELC	Linear Combination of Reactions

EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
ESI	Specific Ion Interaction Theory
MCE	Cation exchange
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MIX	Ion exchange
MSL	Solubility
MTD	Temperature difference

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